

Feature-Matching Technique

Performance Report

5 correspondence-finding techniques, 1 pipeline, measured head-to-head

Held constant across every run

- Bowl-shaped projection surface, gain correction, ego-vehicle self-mask, bilateral-filter denoising
- The same RANSAC similarity-transform fit (rotation + scale + shift only, no skew), applied to the query grid — identical downstream logic for every technique

What varied — the correspondence-finding technique

-  Template matching
-  SIFT
-  ECC (tilted)
-  Phase correlation
-  LoFTR

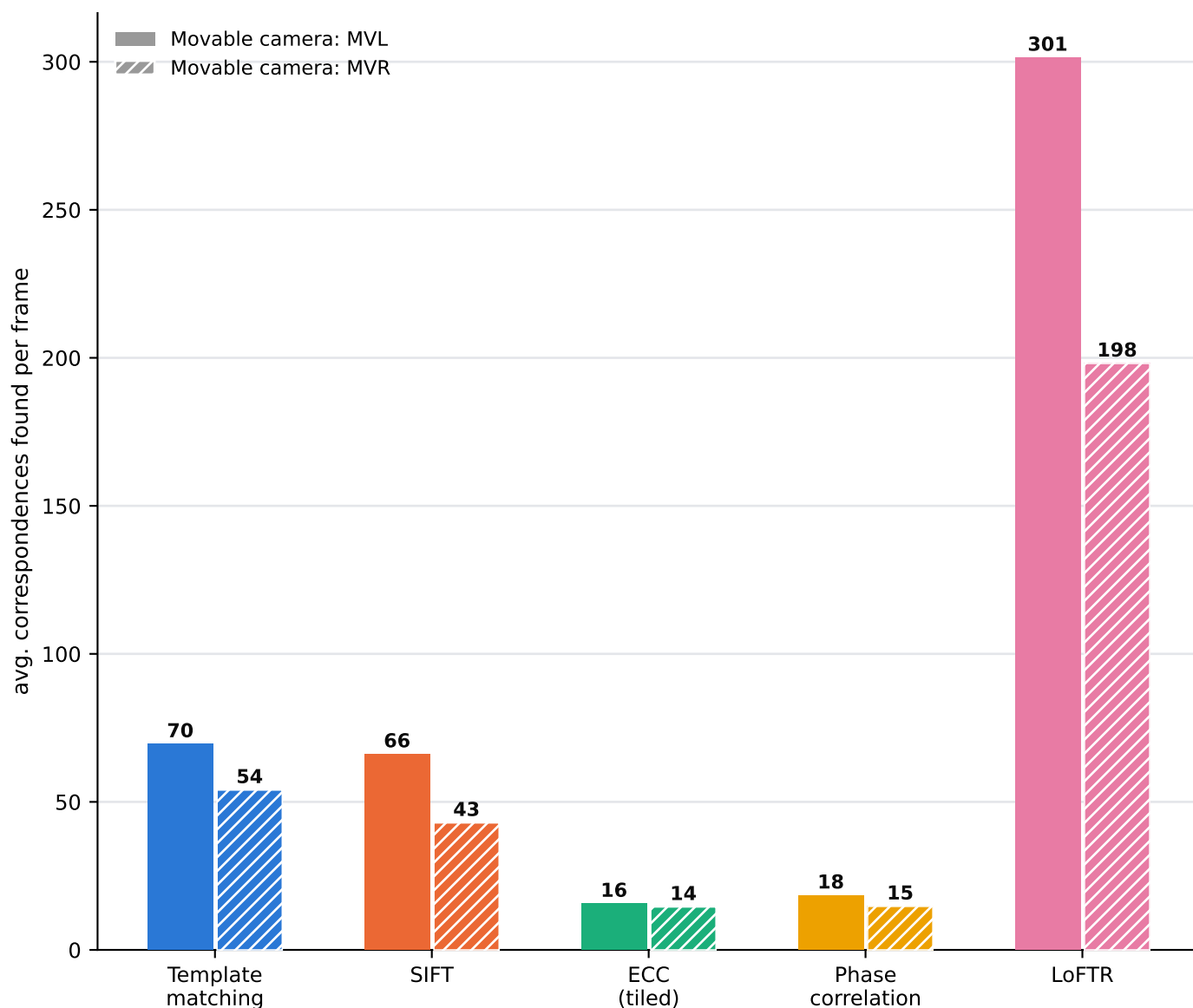
Test set & metric

- 10 sandbox frames × 4 camera-overlap pairs each (FV-MVL, FV-MVR, RV-MVL, RV-MVR)
- Metric: seam-disagreement score (mean pixel difference where two cameras render the same ground) — reported as % improvement from before to after correction, lower score is better, averaged over the 10 frames

RESULT 1

How Many Correspondences Did Each Find?

Averaged across all 10 frames — the raw amount of signal each technique had to work with



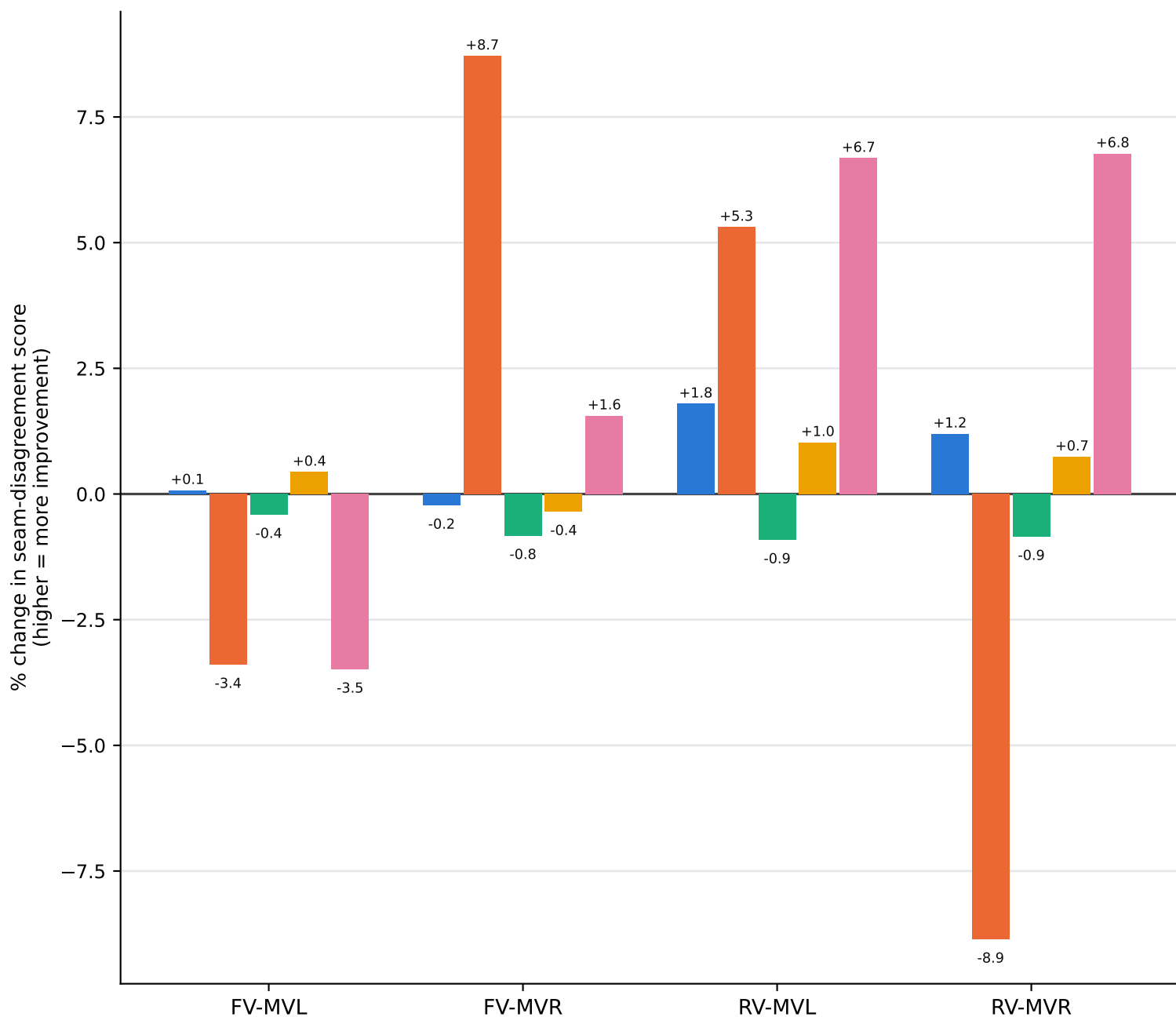
LoFTR finds 4-6x more correspondences than every other technique — the expected result, since it's specifically built for exactly this failure mode (low-texture, repetitive patterns like asphalt) that starves classical detectors.

ECC and phase correlation are tiled (many small patches); counted here is how many tiles passed their confidence threshold.

RESULT 2

Did It Actually Reduce Ghosting?

% change in seam-disagreement score per camera pair — positive = less ghosting, negative = worse

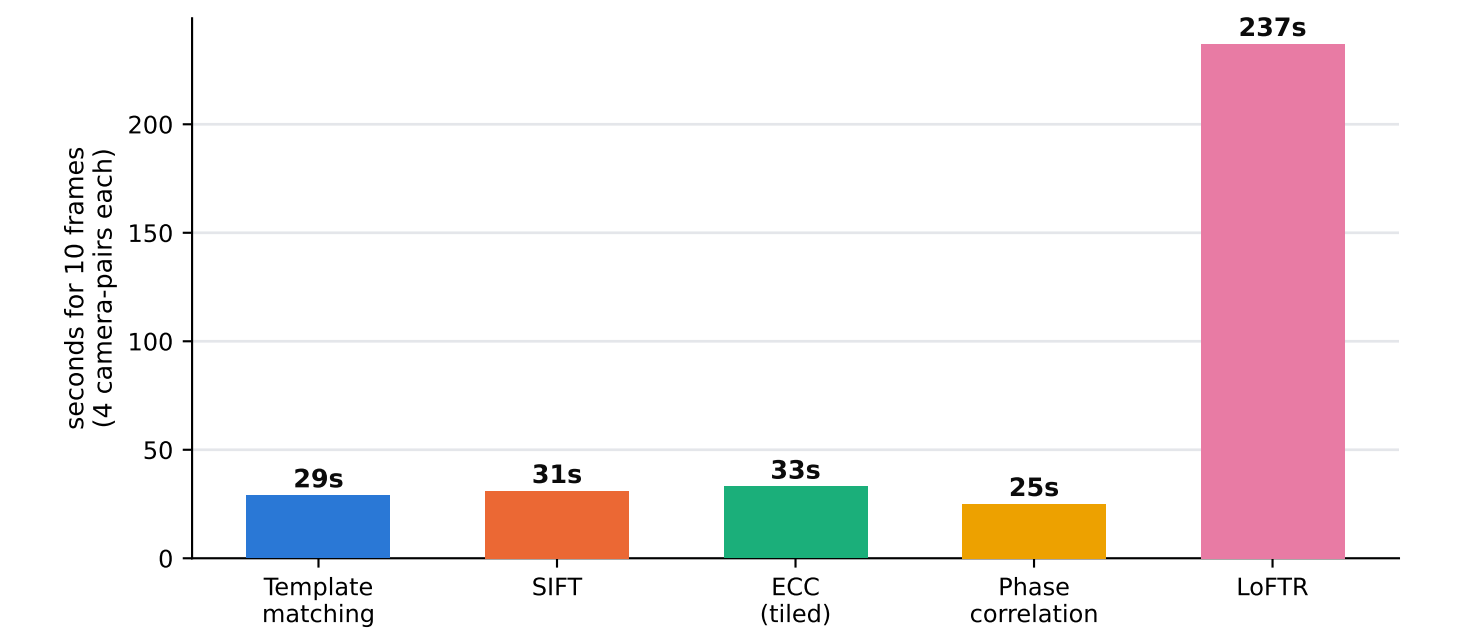


LoFTR is the only technique with a consistent positive trend (3 of 4 pairs improved, up to +6.8%). SIFT's average looks similar to baseline, but that hides high variance — it helps 2 pairs and hurts 2, swinging $\pm 9\%$.

RESULT 3

Timing, and the Overall Ranking

10 frames, 4 camera-pairs each — everything else identical



Overall ranking (avg improvement across all 4 pairs)

Technique		Avg %	Std dev	Verdict
	LoFTR	+2.88%	4.24	Best — real, consistent gain
	Template matching	+0.71%	0.82	Baseline — no real effect
	Phase correlation	+0.46%	0.52	No real effect
	SIFT	+0.44%	6.96	Unstable — cancels out
	ECC (tiled)	-0.76%	0.20	Net negative

LoFTR runs on CPU here (no GPU available) — the 237s/10-frames cost would shrink substantially on a GPU.

What Happened With Each Technique, And Why

LoFTR

Winner. Its correspondence count (300/198) dwarfs everything else because it's a detector-free deep model trained specifically to find matches in low-texture, repetitive regions — exactly what asphalt is. More (good) correspondences means the RANSAC similarity fit has far more to work with, producing a more stable, better-conditioned correction. Visually confirmed on a railing spanning the RV-MVL seam: noticeably smoother, less kinked than the baseline. Cost: ~8x slower than the others on CPU.

SIFT

A real step up in match quality over ORB (which found almost nothing on this asphalt in the very first attempt) — SIFT found a comparable count to the existing NCC method. But the average hides the real story: results swing from +8.7% to -8.9% across different camera pairs. Classical sparse detectors on repetitive texture (aggregate grain, lane-paint patterns) can produce confidently wrong matches, not just too few of them — RANSAC can't always filter that out.

ECC (tiled)

Failed cleanly, and instructively. Tried first on the whole overlap region: didn't converge at all (too much of the region is parallax-affected content, not matching ground). Tiled down to 90x90px patches: most tiles still failed to converge, and the few that did had low confidence. Whole/large-region pixel correlation is a poor fit when the region mixes real matching ground with parallax-shifted objects.

Phase correlation

Same root problem as ECC, different mechanism. FFT-based phase correlation assumes the two patches are nearly identical up to a pure shift; measured confidence scores here topped out around 0.2 (out of 1.0) even after tiling, versus the ≥ 0.45 threshold needed to trust a match. Essentially no usable signal from this technique on this data.

Template matching

"Verdict" in the ranking table reflects net effect across the 4 pairs, not raw correspondence count.

RECOMMENDATION

What To Actually Do With This

If you want to swap the seam-alignment technique for the full 542-frame batch: use LoFTR. It's the only technique that produced a real, consistent improvement (up to +6.8% on individual pairs, +2.88% average) rather than noise-level change or an unstable swing between helping and hurting.

But weigh the cost:

- ~8x slower than every other technique tested (237s vs 25-33s for 10 frames on CPU)
- At that rate, the full 542-frame batch would take roughly 3-4 hours on this machine
- Requires the PyTorch + kornia dependency (already installed in .venv, ~1.4GB)
- The gain, while real, is modest in absolute terms (a few percent) — it does not touch the height-parallax smearing, which remains the dominant visible artifact

Practical suggestion: given the modest absolute gain and the ~8x runtime cost, LoFTR is worth using if geometric seam quality specifically matters for your next step (e.g. training data curation). If you're optimizing for throughput instead, the existing dense-NCC template matching remains a reasonable default — it costs almost nothing extra and the seam quality it leaves behind is already close to calibration-optimal, per the earlier finding that the true miscalibration is tiny.

End of report.